

AI ADOPTION TRACKER

August 2026: Adoption Rises to 22.4%

- We update our AI adoption tracker through August 2026.
- **AI-related investment growth** remains strong, particularly for semiconductors, where equity analysts expect global revenue to reach \$863bn (ann.) by the end of 2026. 2026 gross profit forecast revisions since 2022Q3 for AI-exposed global public companies currently stands at \$1.1tn. AI-related hardware investment in the US national accounts also continues to rise and now stands at \$485bn (3-month annualized average; 1.5% of GDP) above its 2022 level. AI-related hardware shipments from Taiwan have grown by 76% since 2022 and US AI hardware net imports ticked up to \$55.9bn in June.
- **AI adoption by firms** rose by 0.9pp to 22.4% among US establishments (with adoption expected to rise to 25.9% in the next 6 months) according to the Census Bureau's Business Trends and Outlook Survey (BTOS). Information is leading sector adoption at nearly 50%. At a detailed subsector level, computing and publishing firms continue to lead adoption, and warehousing and miscellaneous manufacturing firms report the largest expected increase in adoption over the next six months. Establishments with over 250 employees continue to lead in adoption (40%; latest six survey average) while adoption has grown more slowly recently among smaller firms. Other business surveys note that 40% of surveyed organizations have begun deploying and scaling AI agents.
- **AI's labor market impact** remains visible but narrow. We continue to observe employment drags in specific industries where AI-use cases have been established such as marketing, graphic design, customer service, and some tech occupations. However, this is offset by construction job growth in data center-exposed categories, which is up 209k since 2022 (relative to broader construction employment trends). The tech share of total employment continues to decline relative to the pre-2022 trend. The correlation between AI adoption and broad labor market slack measures remains limited but has strengthened in recent months. Corporate layoffs attributed to AI affected 11k employees in June, bringing the 2026 YTD total to 113k. 22% of Russell 3000 companies mentioned AI and labor-related keywords in 2026Q2.
- We continue to observe large **impacts on labor productivity** in the limited areas where generative AI has been deployed but only modest signs of economy-wide impacts. Academic studies imply a 23% average uplift to productivity, while company anecdotes imply slightly larger efficiency gains of around 32%.

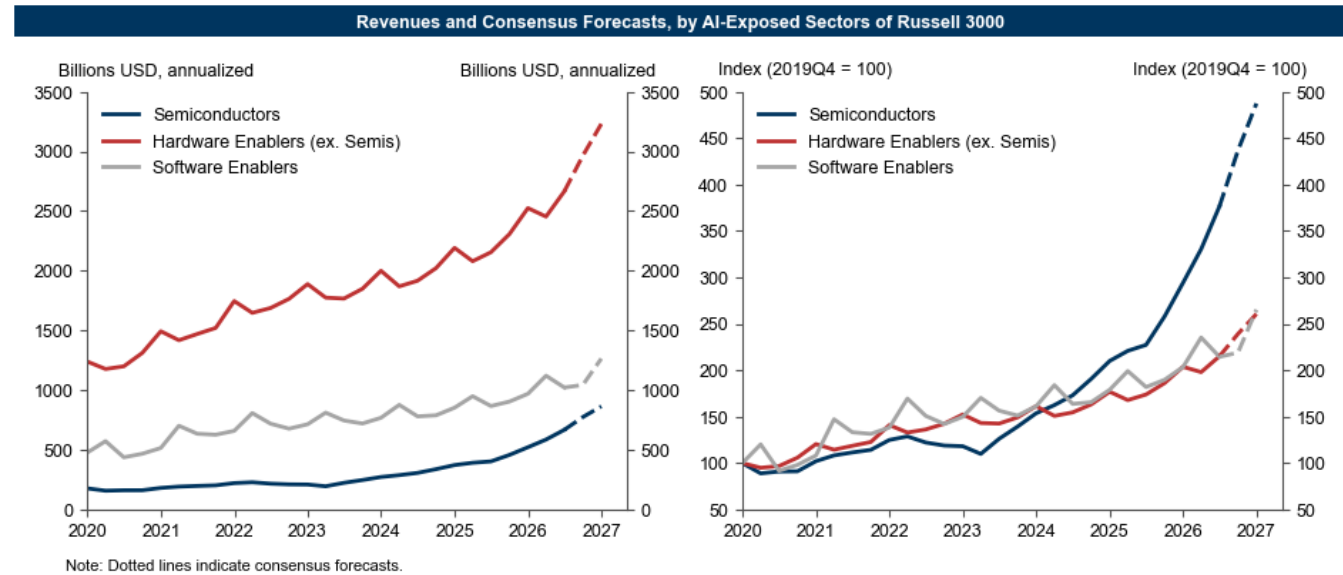
Sarah Dong+1(212)357-9741 | sarah.dong@gs.com
Goldman Sachs & Co. LLC**Joseph Briggs**+1(212)902-2163 | joseph.briggs@gs.com
Goldman Sachs & Co. LLC

Industries with higher AI adoption rates are now showing a slight acceleration in productivity growth over the past year in official US data.

August 2026: Adoption Rises to 22.4%

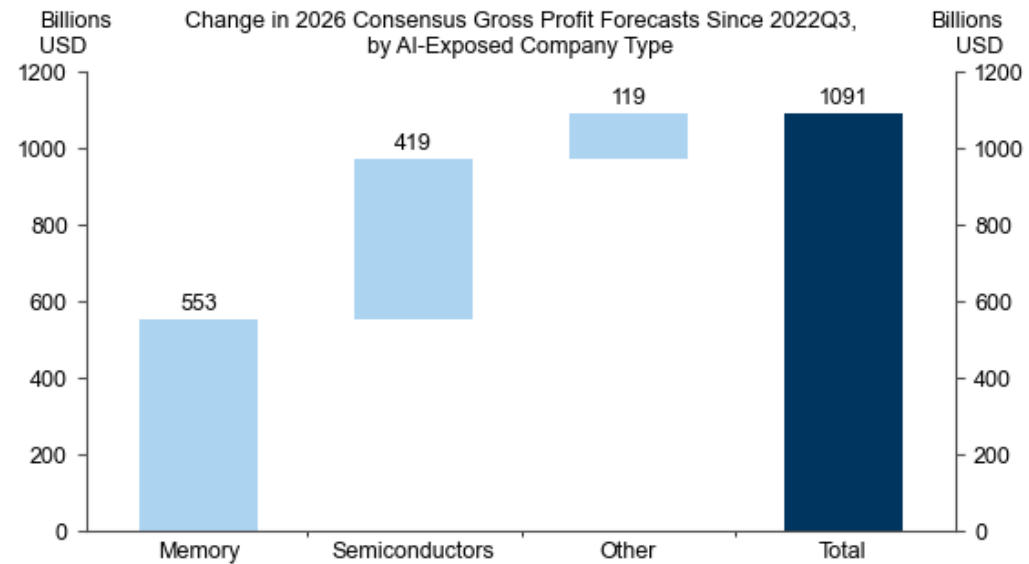
AI Investment

Exhibit 1: Semiconductor Company Revenues Are Projected to Rise to \$863bn (Annualized; vs. 2022) by End-2026



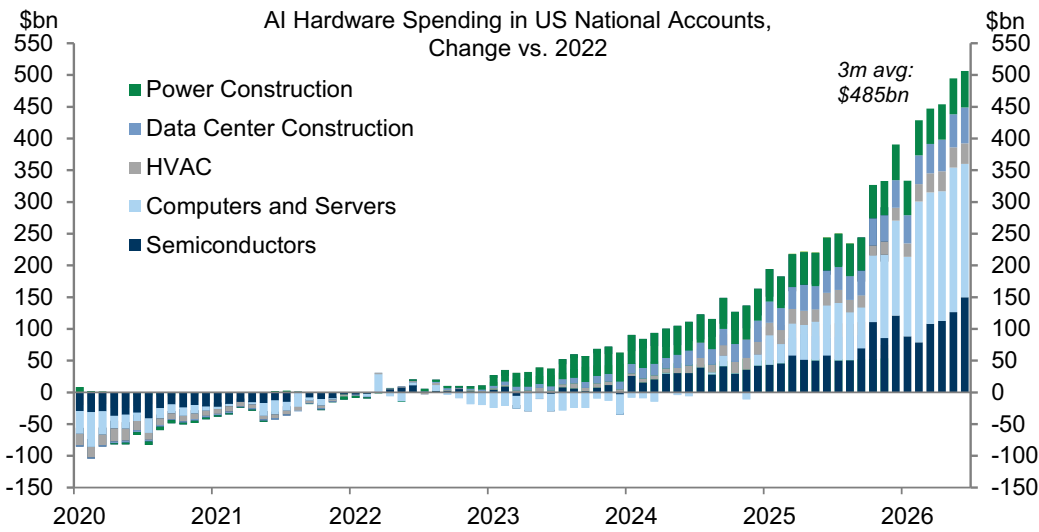
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 2: AI-Exposed Gross Profit Forecast Revisions For 2026 Currently Stand at \$1.1tn



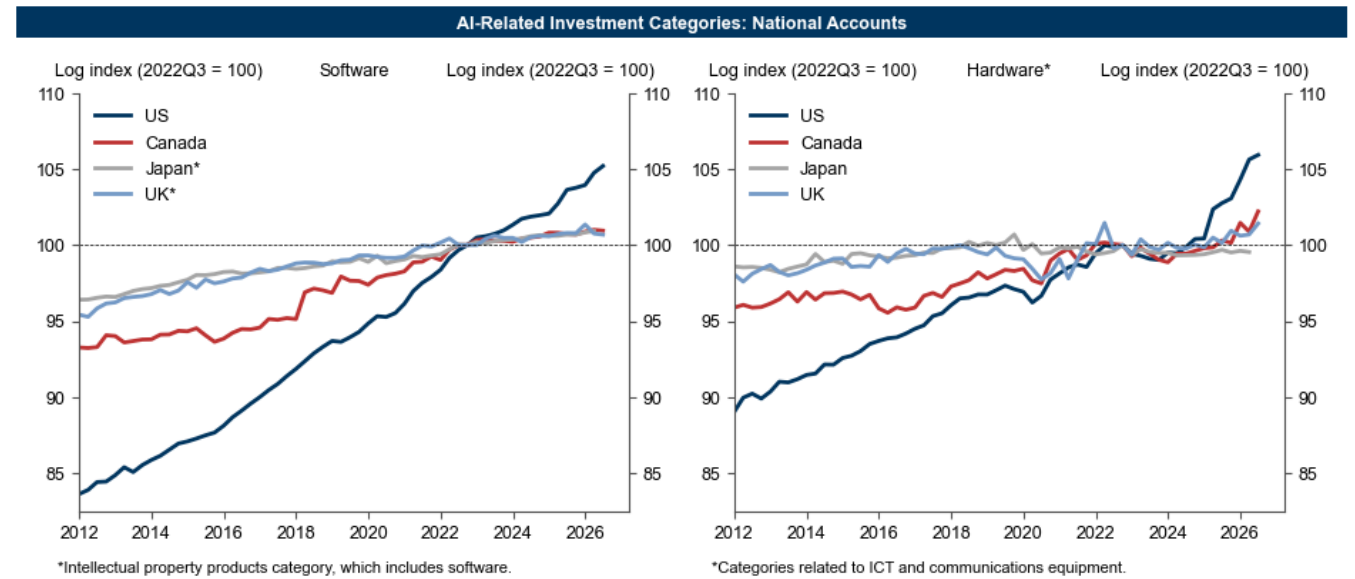
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 3: AI Hardware Investment in the US National Accounts Has Increased by \$485bn Since 2022 on a Three-Month Annualized Basis



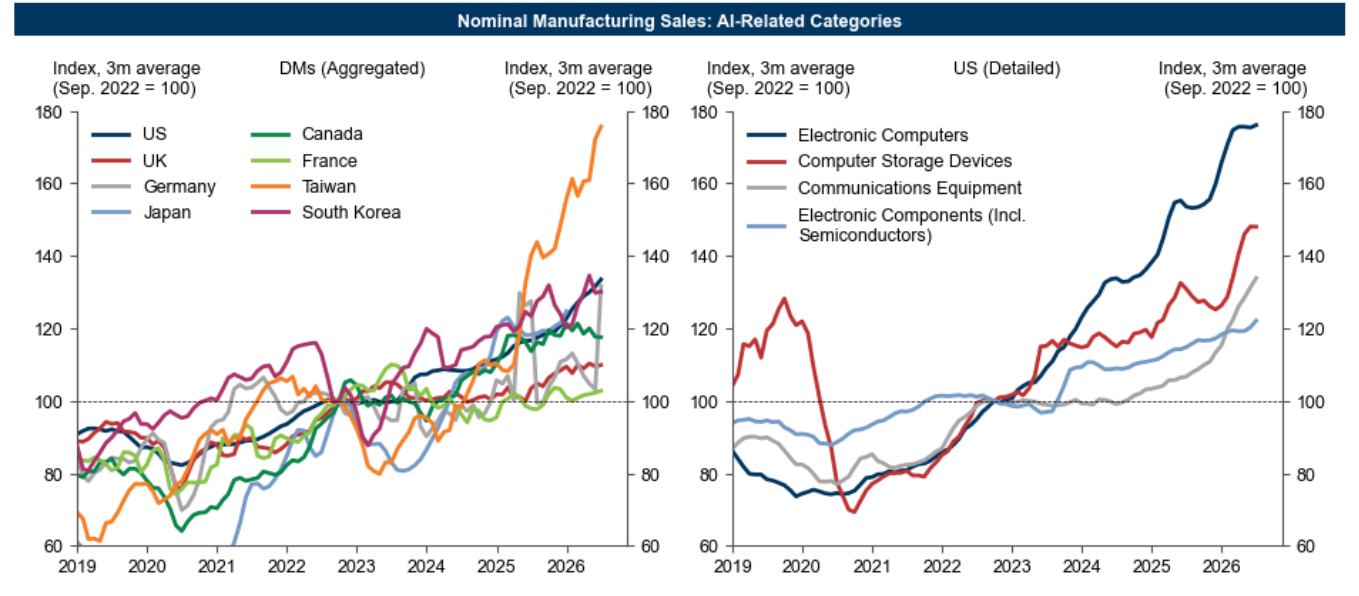
Source: Census Bureau, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: US Investment in AI-Related Hardware and Software Ticked Up in Q2



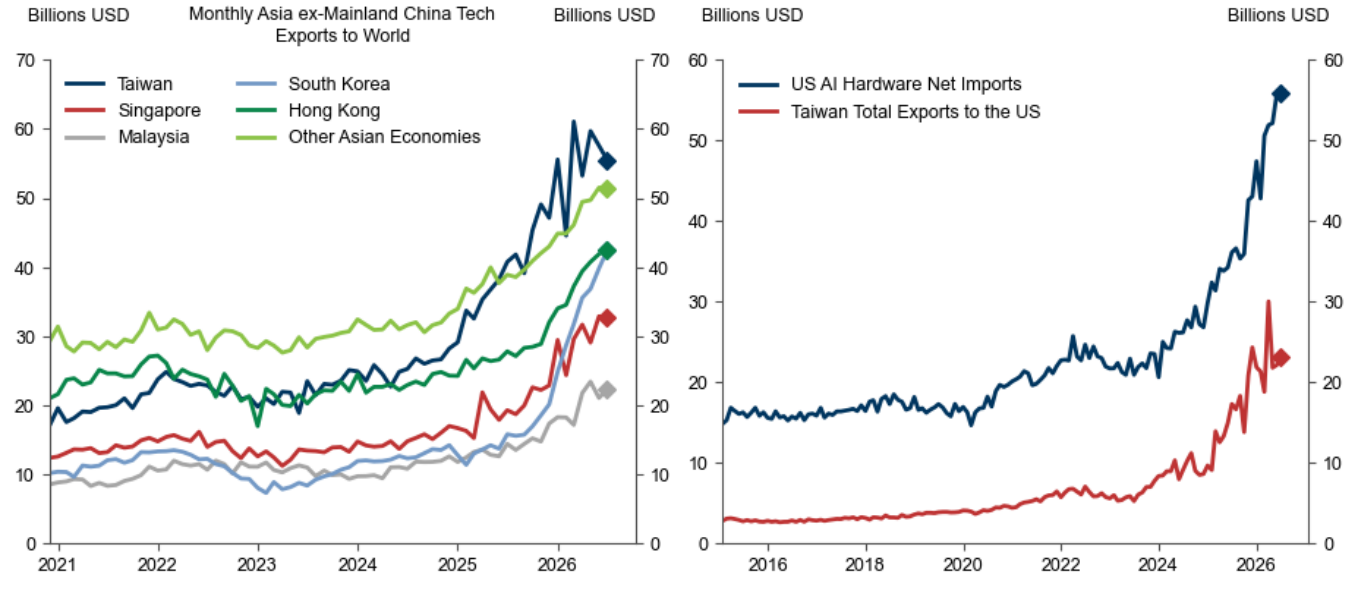
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 5: AI-Related Manufacturing Shipments From Taiwan Have Grown by 76% Since 2022



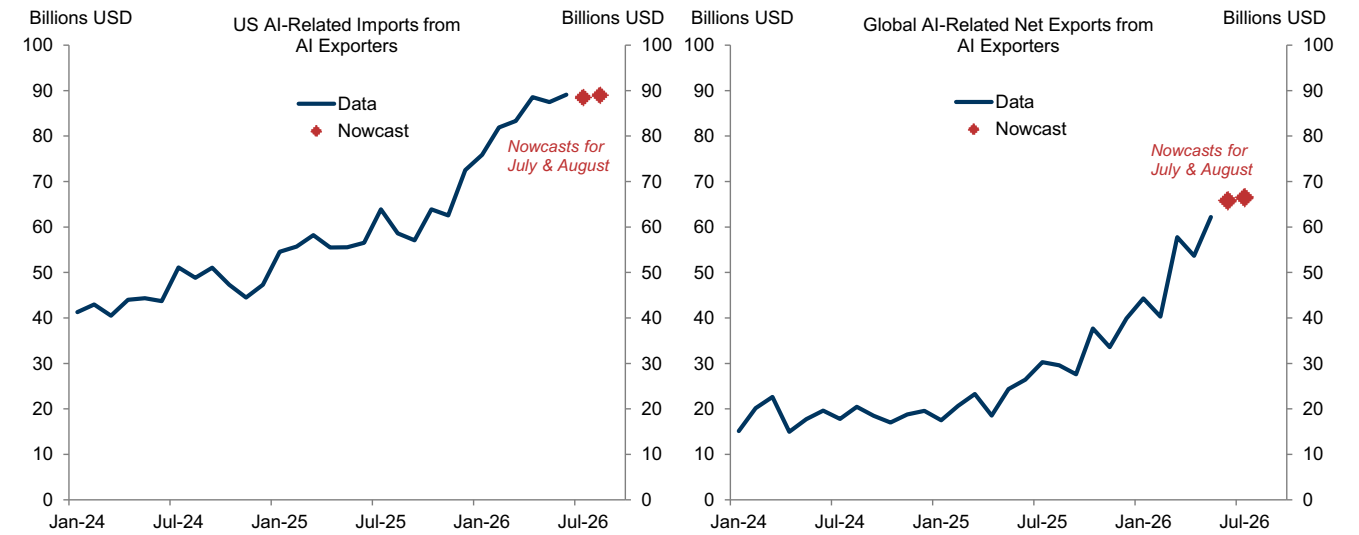
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: US AI Hardware Net Imports Stood at \$55.9bn in June



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 7: We Expect \$66bn in Global AI-Related Net Exports from AI Exporters in August

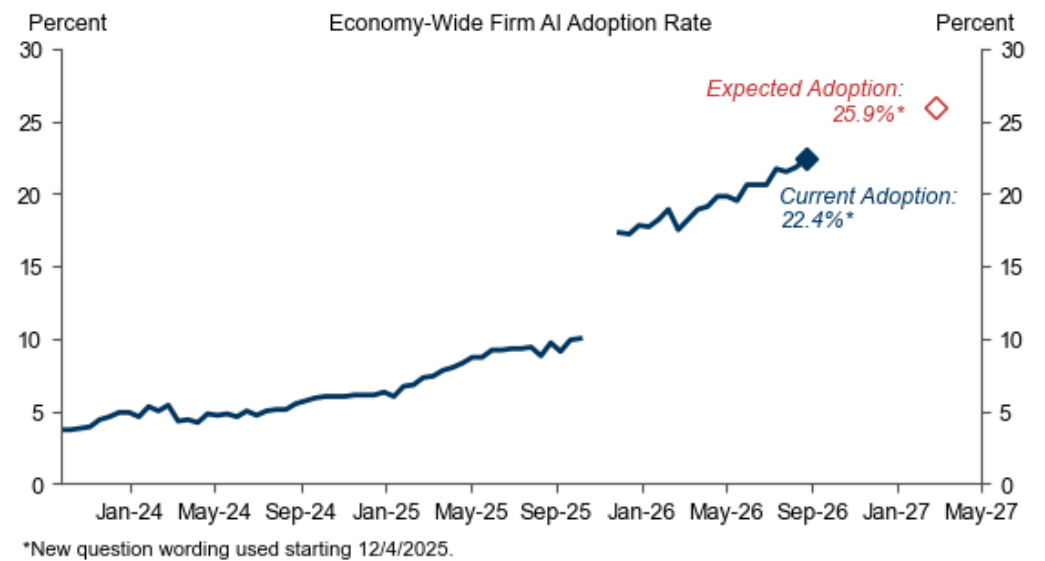


Source: Goldman Sachs Global Investment Research

AI Adoption by Firms

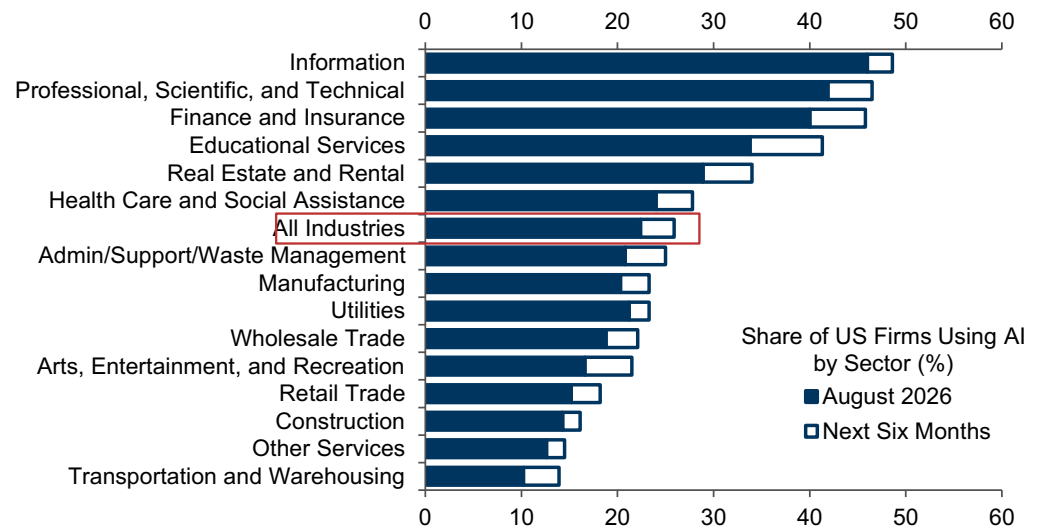
Note: The Census Bureau has rephrased questions from asking about AI use in “producing goods and services” to asking about AI use in “any business functions” starting from the December 4, 2025 release. There was a level shift observed in conjunction with the new question wording, and we will be tracking adoption rates under the new methodology going forward.

Exhibit 8: 22.4% of Firms Are Using Artificial Intelligence in Their Regular Business Functions (+0.9pp From July); 25.9% Expect to Use AI In the Next 6 Months



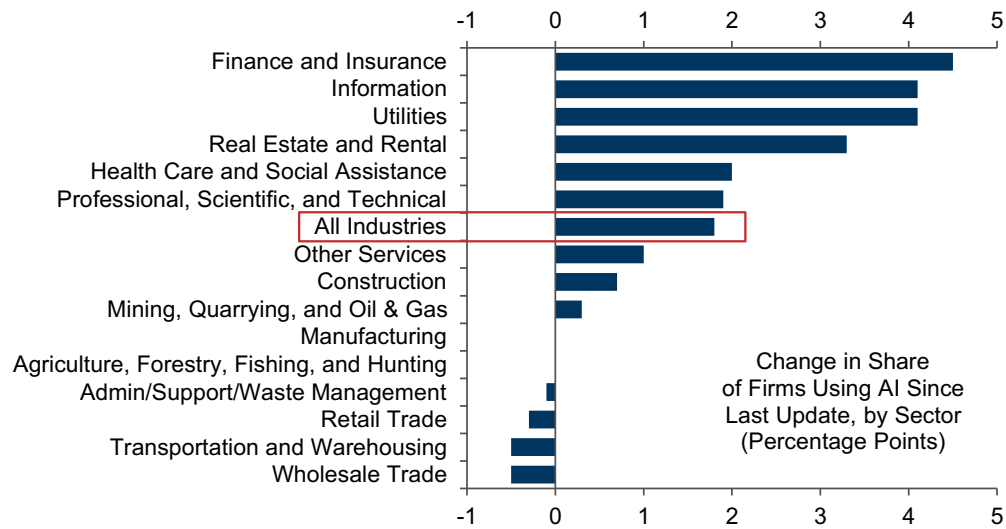
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 9: Adoption Is Approaching 50% in the Information Sector



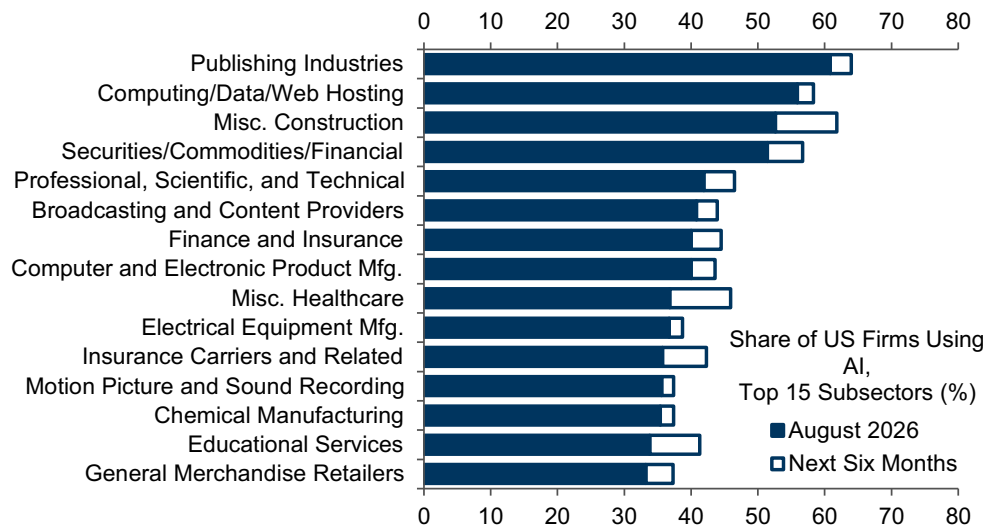
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 10: Finance and Insurance Firms Had the Largest Gains in Adoption Since Our Last Update



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 11: Publishing and Computing Continue to Lead Adoption at the Subsector Level



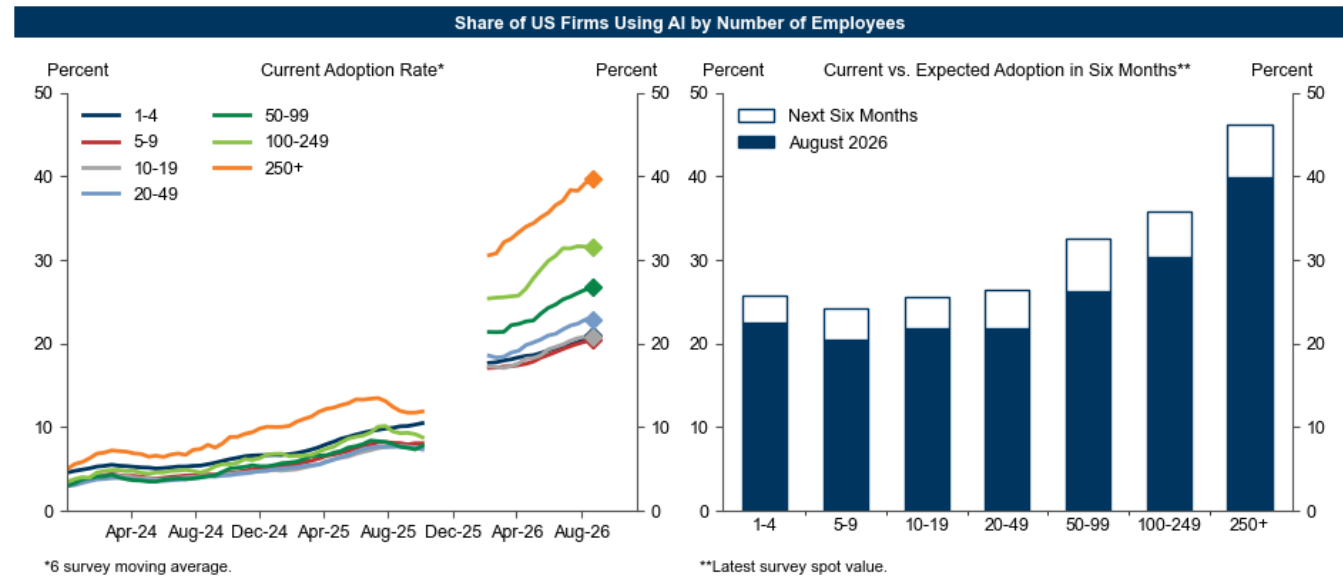
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 12: Warehousing and Storage and Miscellaneous Manufacturing Firms Report the Largest Expected Adoption Increases in the Next 6 Months



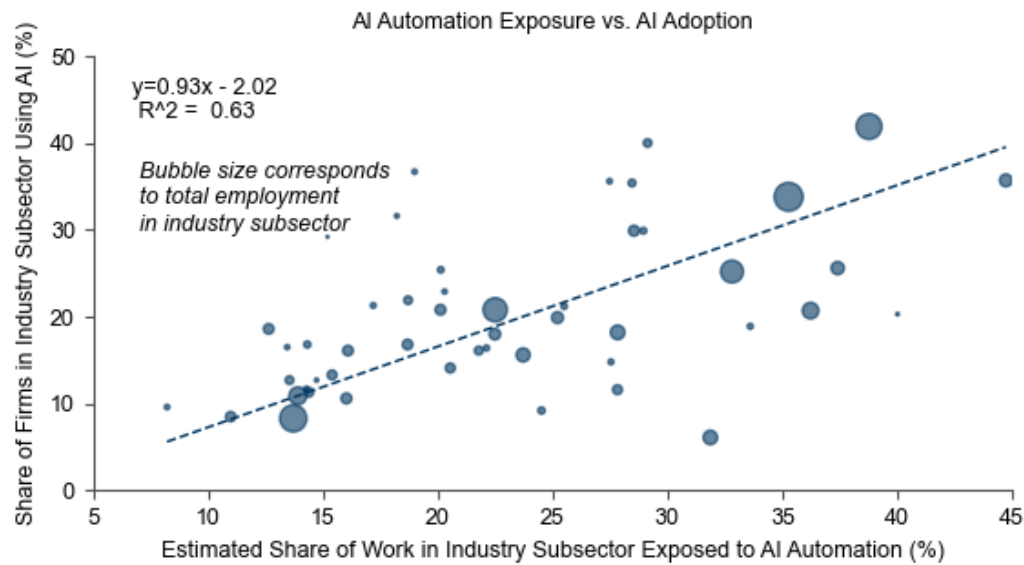
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 13: 40% of Large Firms With 250+ Employees Expect to Be Using AI in Six Months



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 14: AI Adoption Remains Strongly Correlated With Our Subsector AI Automation Exposure Scores



Share of work exposed to AI automation is a GS estimate based on the methodology in “Global Economics Analyst: The Potentially Large Effects of Artificial Intelligence on Economic Growth” (March 26, 2023).

Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 15: 40% of Surveyed Organizations Have Begun Deploying and Scaling AI Agents

Source	Date	Current Adoption / Impacts	Future Adoption / Impacts
McKinsey	August 2026	40% of large organization respondents have adopted AI agents and are scaling their deployment throughout the organization, up from 27% last year. Nearly a third of respondents reported that they had used agentic coding tools to build software in-house instead of purchasing it. 80% of respondents said AI has improved their individual productivity.	60% of surveyed organizations expect to increase their AI investments in the year ahead. 39% of respondents expect AI-related declines in their organization's total employment in the coming year.
Gallup	July 2026	47% of surveyed US employees in Q2 said their organization has integrated AI tools to improve workplace productivity, up from 41% last quarter. Individual AI use at work also increased to 52% of respondents. Among AI users, the most common uses of AI are writing and editing (51%), search or research (49%), and general assistance or problem-solving (39%).	The highest productivity ratings come from employees using AI for coding assistance and automation or process automation. Employees who use AI for a greater variety of tasks were also more likely to report that AI positively affects their personal productivity. Although corporate adoption and AI awareness has grown notably, one in five employees remain unsure if AI has been integrated at their organization.
WRITER	April 2026	52% of employees say they've used AI agents over the past year. Employees reported saving around 6 hours per week with AI tools, while executives reported saving twice as much time at nearly 12 hours per week. The top use cases include editing and proofreading, personalized communication, and data analysis and insights. Few surveyed leaders reported seeing significant ROI from generative AI (29%) or AI agents (23%).	69% of executives report doing layoffs due to AI, but 39% admit their company doesn't have a formal strategy to drive revenue from AI.
Harvard Business Review and Apian	April 2026	Most organizations have moved beyond the experimentation phase with AI and 58% of surveyed organizations have AI in production. 69% of respondents reported that legacy systems are limiting their organization's ability to scale AI. Most respondents are seeing some returns from AI investments, but only 16% report realizing a high degree of measurable value.	Many organizations are currently exploring agentic AI systems (under consideration by 62% of respondents and currently used by 25%). Although nearly all organizations agree that clear guardrails are needed for AI agent adoption, fewer than half say their organization has defined such rules.
Gallup	April 2026	Half of all surveyed adults reported using AI in their jobs and 27% of employees in AI-adopting organizations say that their workplace has changed in disruptive ways due to AI in the past year. 65% of adopters said that AI has improved their productivity and efficiency, regardless of how often they personally use AI at work.	Eighteen percent of all U.S. employees say it is very or somewhat likely their job will be eliminated within the next five years due to AI or automation. Among employees working in organizations that have adopted AI, that share rises to 23%. However, employees in AI-adopting organizations were more likely to report both current expansions and reductions in headcount.
NVIDIA	March 2026	64% of respondents said their organizations are actively using AI in their operations. 28% said they're still in the assessment phase, while 8% said they're not using AI and have no plans to. Regional adoption among respondents was 70% in North America, 65% in EMEA, and 63% in APAC. Larger companies (with over 1000 employees) demonstrated broader adoption, deployed more use cases, and reported greater ROI. More than half of respondents (53%) said improved employee productivity was one of the biggest impacts AI had on business operations.	86% of respondents said their AI budgets would increase this year and nearly 40% said budgets would increase by 10% or more. Respondents reported that workflow and operations and retaining the right expertise to scale AI solutions were the key challenges to adoption.
Deloitte	January 2026	In the latest survey, 34% of companies were starting to use AI to deeply transform their businesses, 30% were designing key processes around AI, and the remaining 37% were using AI at the surface level with little or no change to underlying business processes. 84% of companies have not redesignated jobs or work tasks around AI capabilities. 25% of surveyed leaders reported that AI is having a transformative effect on their organizations, more than double from 12% a year ago.	Survey results showed that insufficient employee skills were the largest barrier to integrating AI into existing workflows. Nearly three-quarters of surveyed companies expect to be using agentic AI at least moderately in two years.
Gallup	December 2025	The percentage of US employees who reported using AI at work at least a few times a week increased from 19% in 2025Q2 to 23% in Q3. Unsurprisingly, employees working in knowledge-based jobs (such as tech or professional services) were more likely to use AI than those in frontline positions. Most common use cases include using AI to consolidate information or data, idea generation, and automating basic tasks. Users most often use AI chatbots and virtual assistants.	Gallup research has shown that employee AI adoption is strongly associated with having greater managerial support for AI and strategic integration of AI in their role, indicating that a strategic people-first approach is needed for effective AI adoption.

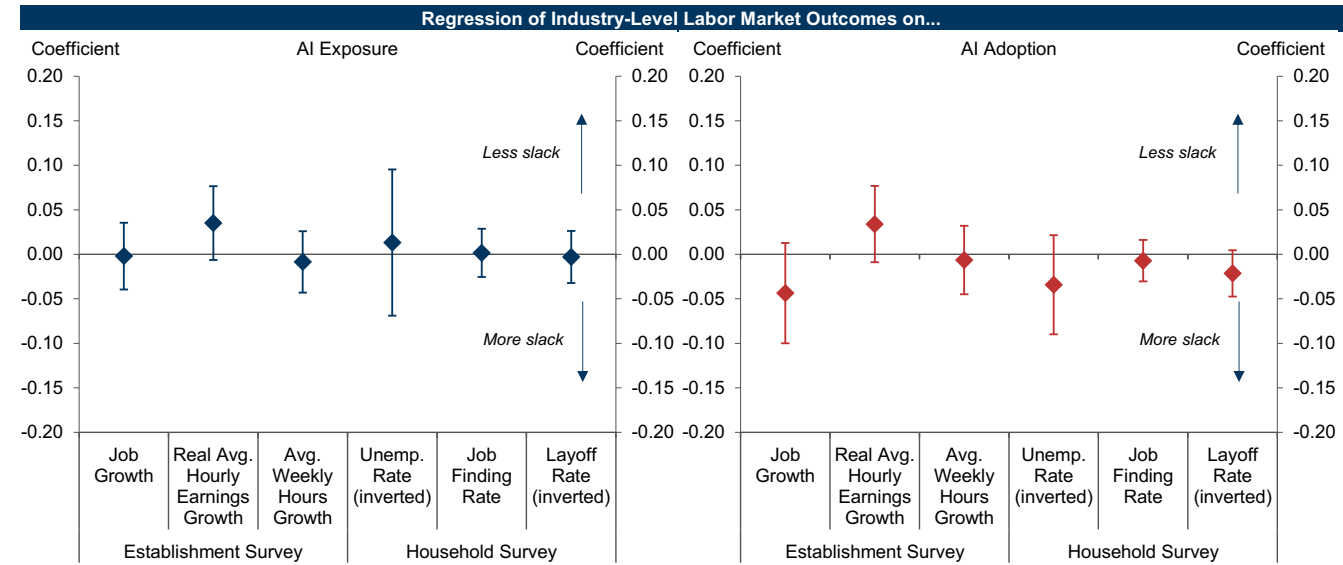
Green shading indicates surveys added since the last update.

Source: Goldman Sachs Global Investment Research

AI’s Impact on the Labor Market

Labor Outcomes

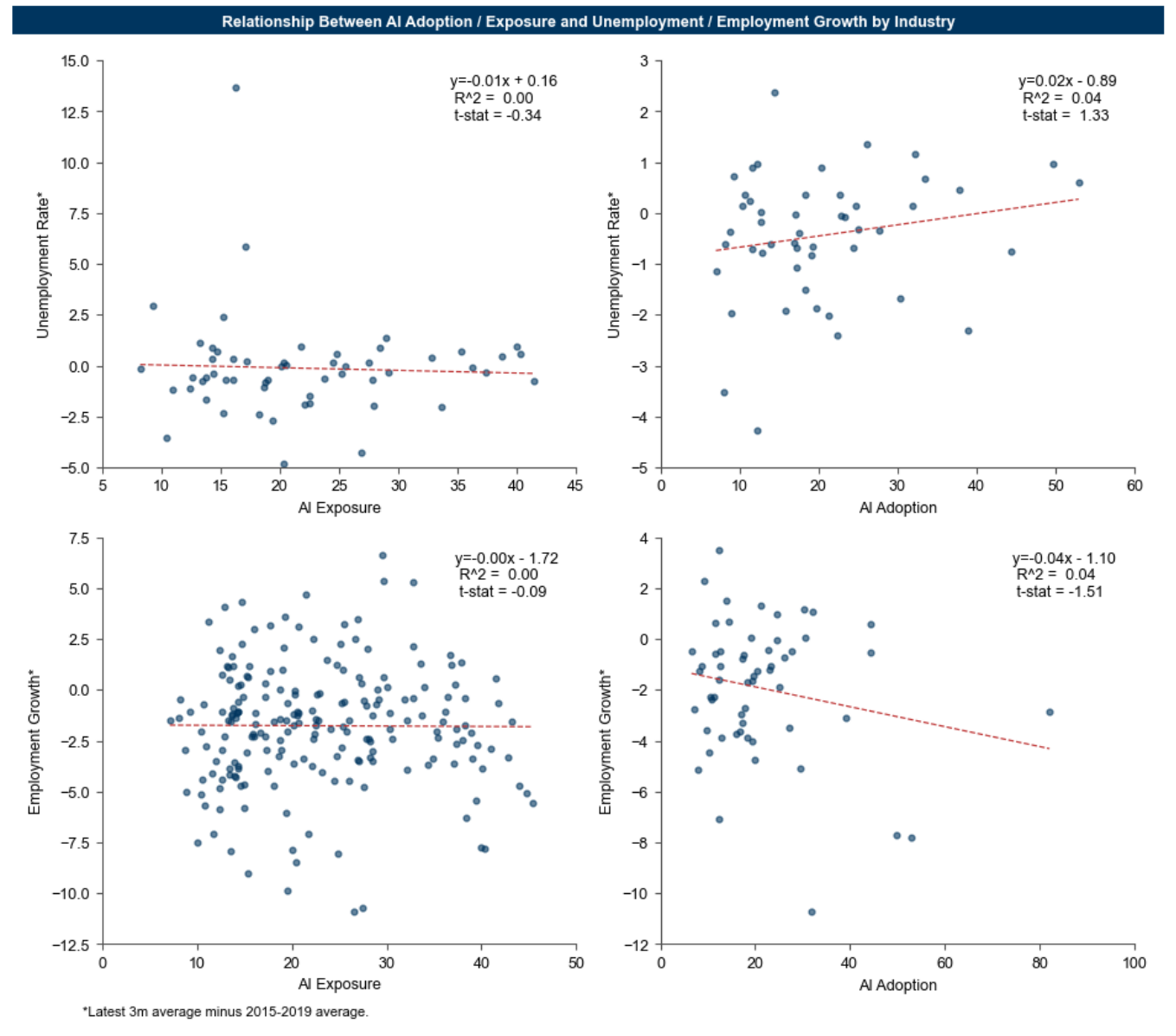
Exhibit 16: Some Signs of a Negative Relationship Between AI Adoption and Labor Market Slack Indicators, Particularly Job Growth



Note: All dependent variables are taken as the deviation from 2015–19 averages. Average coefficient over the past three months is plotted.

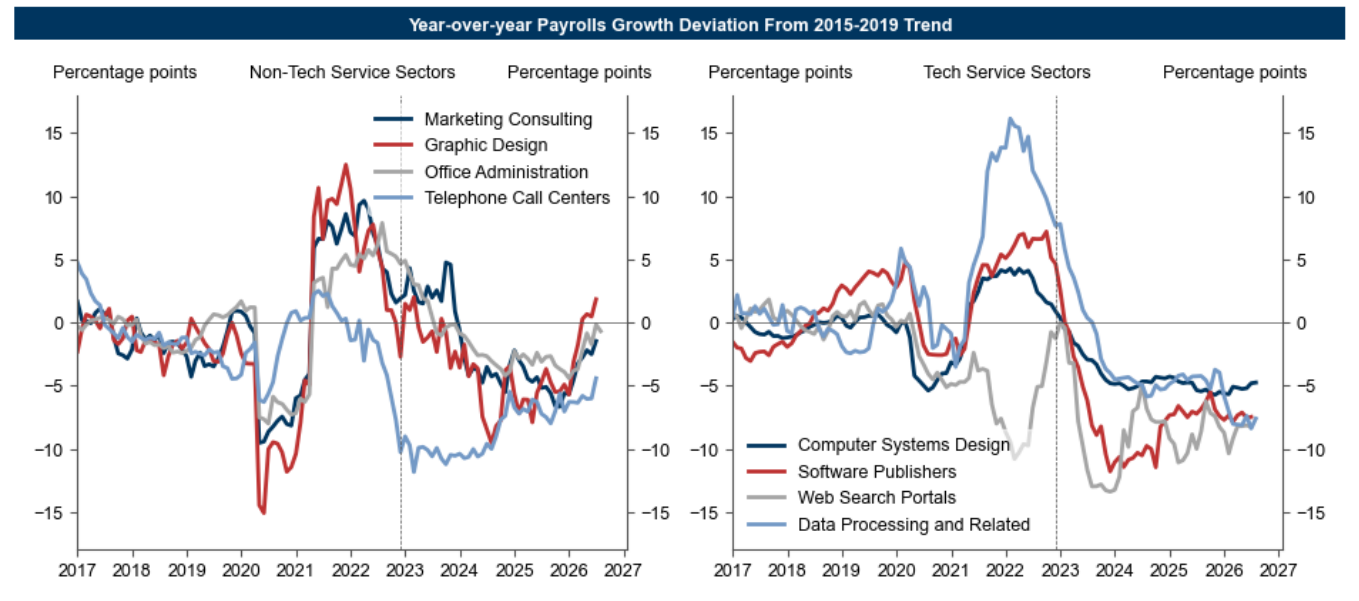
Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

Exhibit 17: Recent Negative Relationship Between Adoption and Employment Growth, but Still Not Statistically Significant



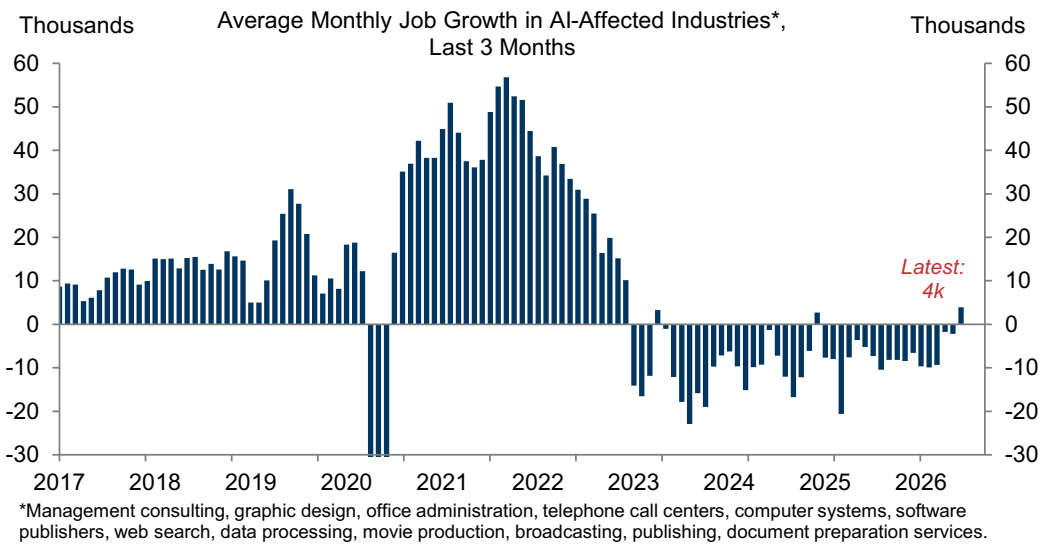
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 18: Employment Growth Continues to Underperform in Occupations (Specifically Tech) Where Anecdotes Suggest AI Is Having an Impact, but Employment Has Recovered Somewhat in Several Affected Non-Tech Sectors



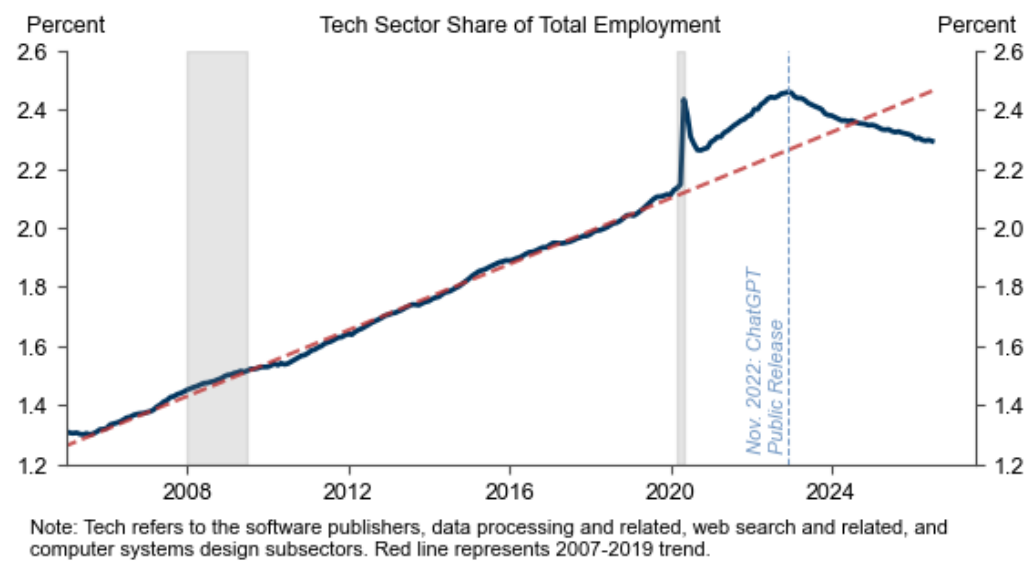
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 19: Job Growth Drag in AI-Affected Industries Has Reversed For Now



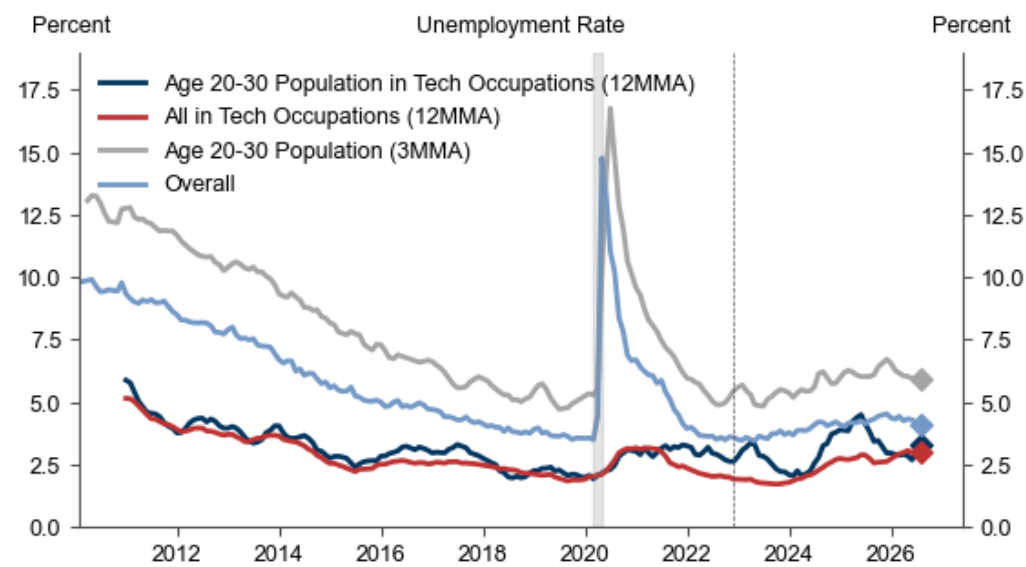
Source: US Bureau of Labor Statistics, Goldman Sachs Global Investment Research

Exhibit 20: Tech’s Employment Share Continues to Decline Below Its Long-run Trend



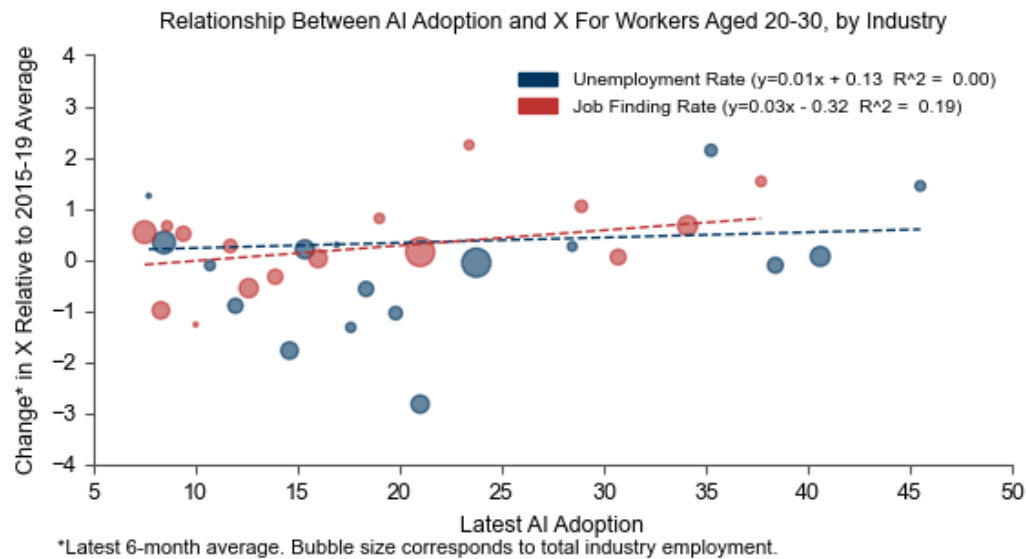
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 21: Unemployment Among Young Tech Workers Has Fallen Back in Line With Unemployment for All Tech Workers



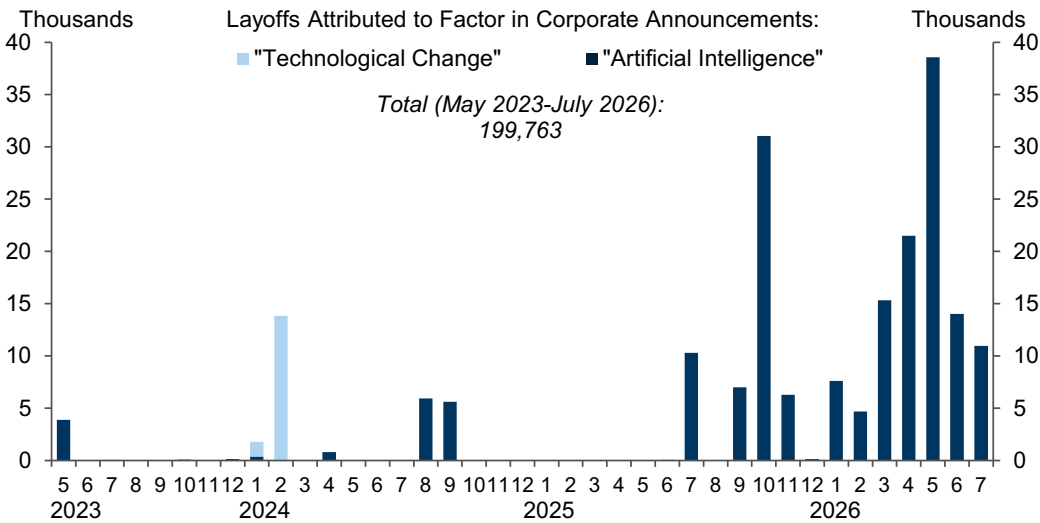
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 22: We Continue to Observe a Slight Positive Correlation Between AI Adoption and Unemployment Among Younger Workers



Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

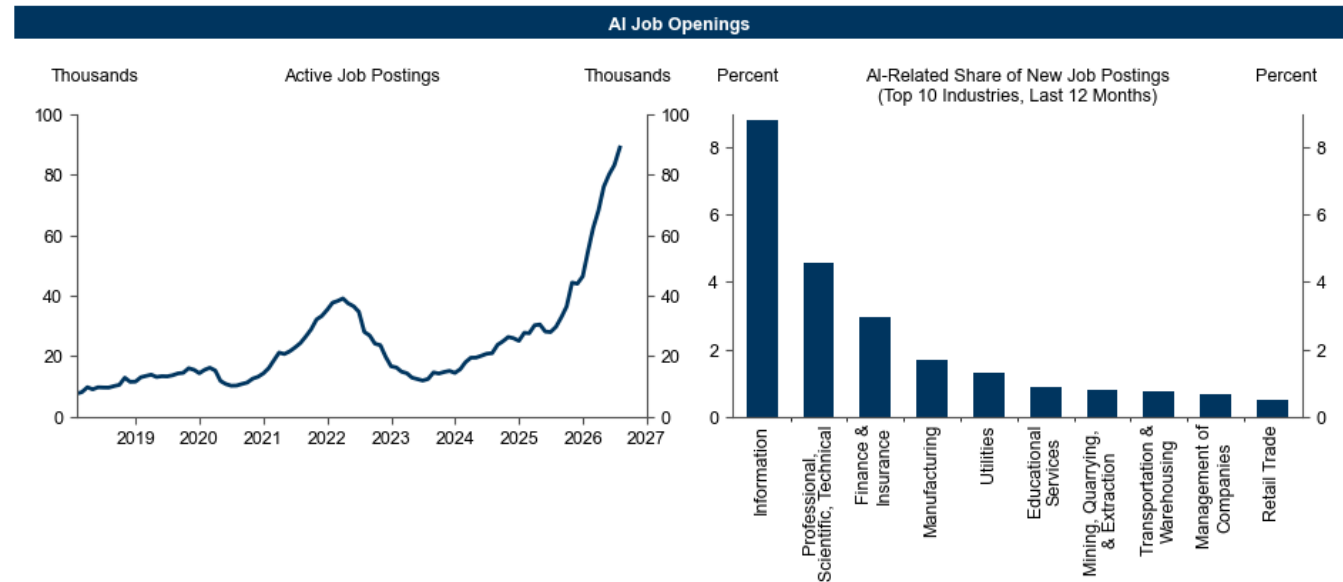
Exhibit 23: AI Was Cited in Layoff Announcements Affecting 11k Employees in July



Source: Challenger, Gray & Christmas, Goldman Sachs Global Investment Research

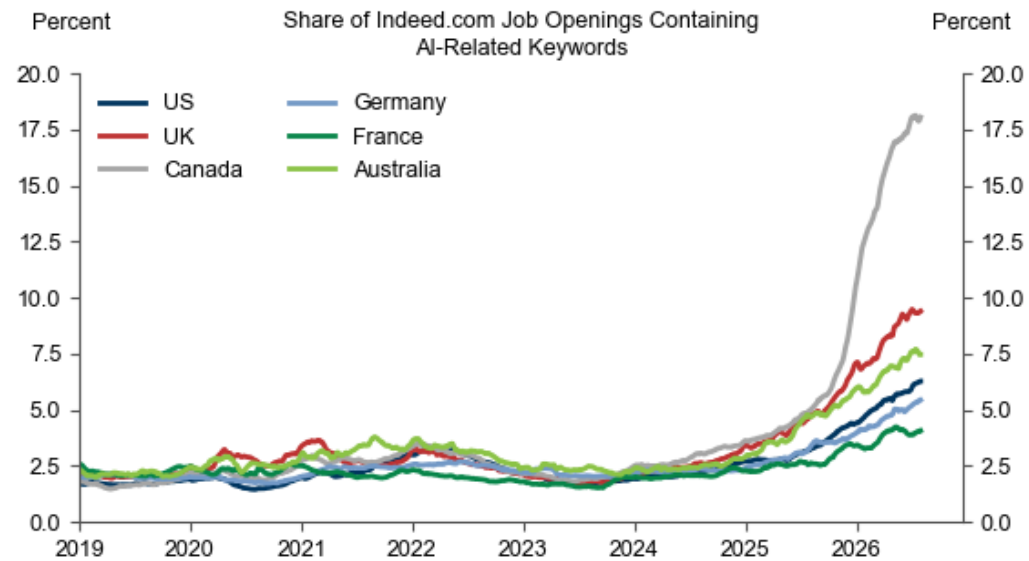
Labor Demand

Exhibit 24: Active AI-Related Job Postings Now Stand at 89k in the US



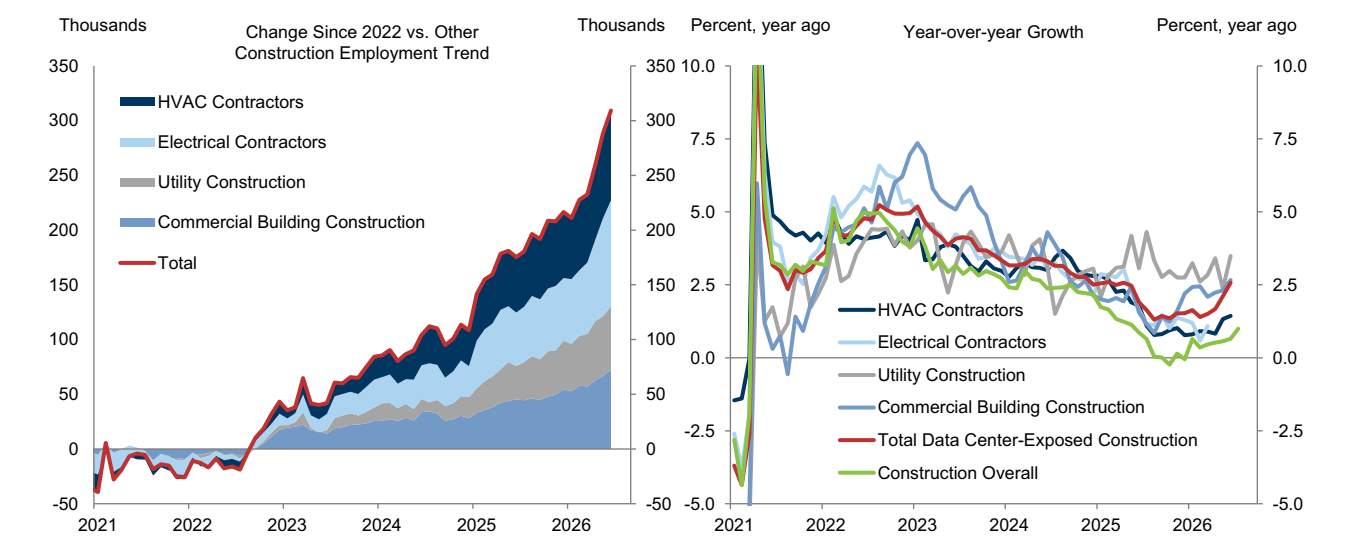
Source: University of Maryland, LinkUp, Goldman Sachs Global Investment Research

Exhibit 25: Share of AI-Related Job Postings Continues to Rise Across All Major DMs



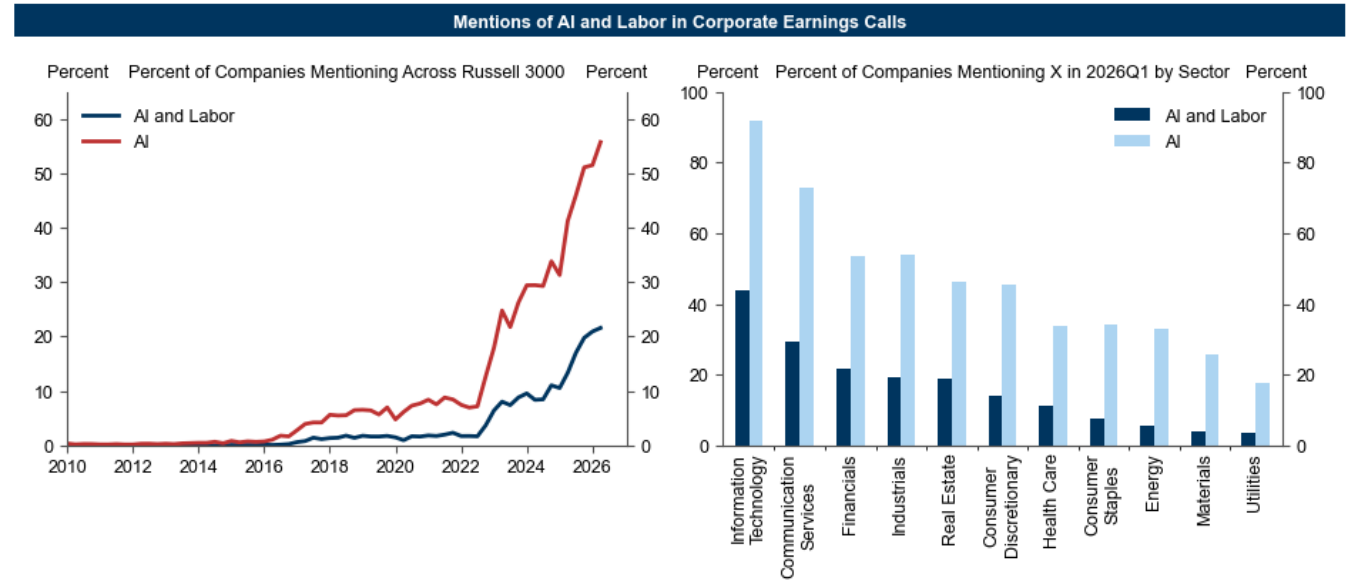
Source: Indeed Hiring Lab, Goldman Sachs Global Investment Research

Exhibit 26: Construction Employment in Data Center-Exposed Categories Has Increased by +309k Since 2022, With a Sharp Acceleration in Recent Months



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

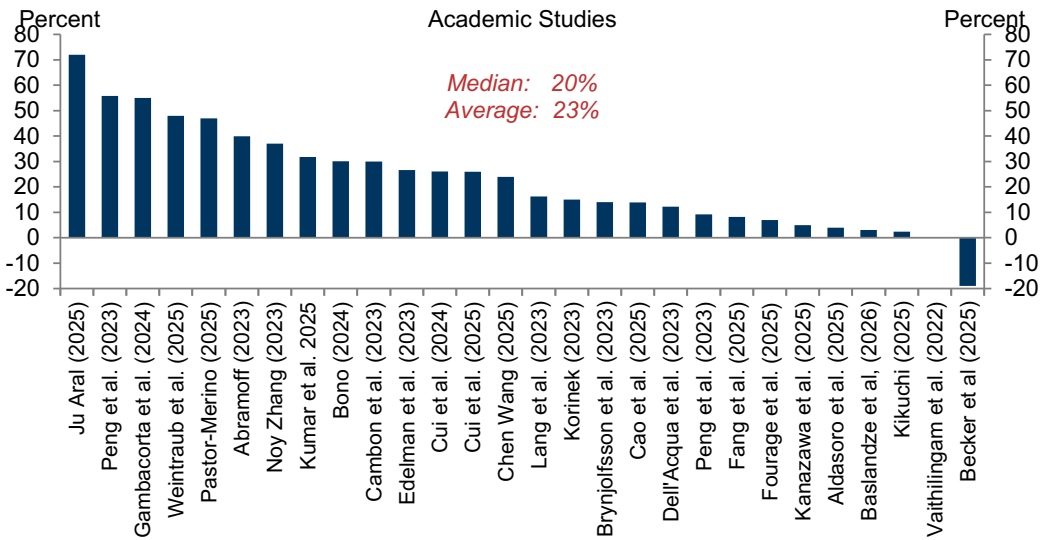
Exhibit 27: 22% of Russell 3000 Companies Mentioned AI and Labor-Related Keywords in 2026Q2 Earnings Calls



Source: GS Data Works, Goldman Sachs Global Investment Research

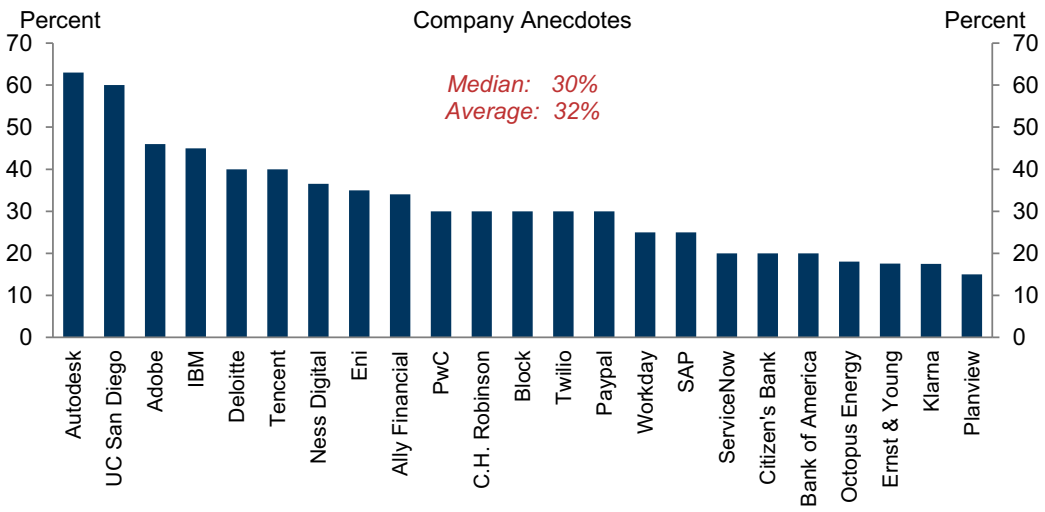
AI’s Impact on Productivity

Exhibit 28: Evidence From Academic Studies Suggests AI Adoption Boosts Labor Productivity by Around 23% on Average...



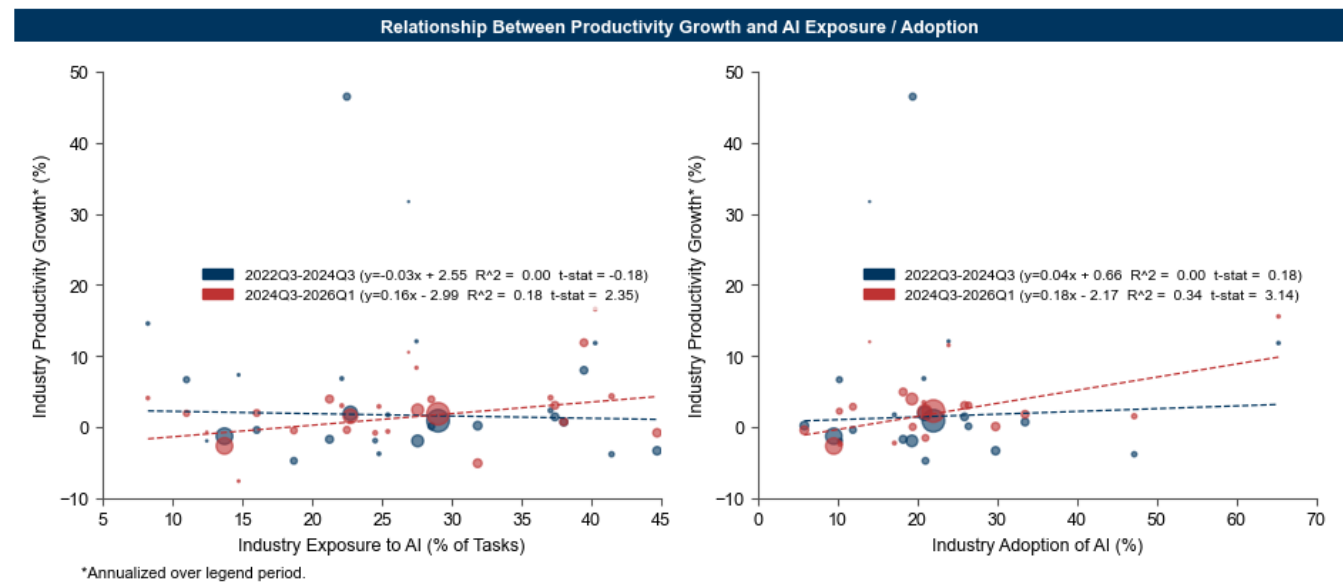
Source: Goldman Sachs Global Investment Research

Exhibit 29: ... And Company Anecdotes Suggest an Average Productivity Boost of 32%



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

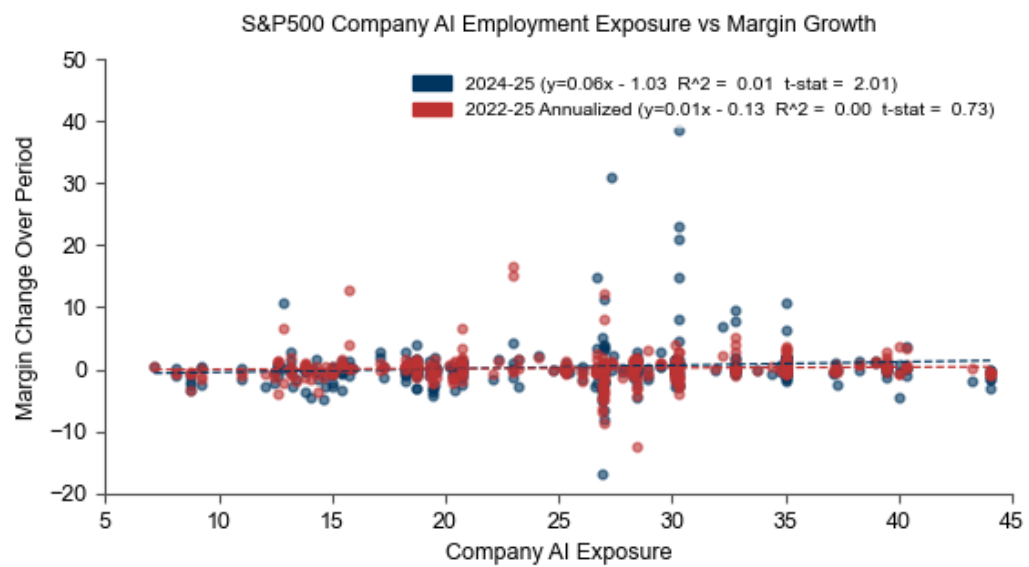
Exhibit 30: Industries with Higher Adoption Rates Continue to Show Slightly Stronger Productivity Growth



Note: Bubble size corresponds to total industry employment.

Source: US Bureau of Labor Statistics, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 31: No Relationship Between Company Level AI Exposure and Margin Growth



Source: FactSet, Goldman Sachs Global Investment Research

Latest GS Research on Artificial Intelligence

- [Global Physical AI: Framing the Forward Progress of Humanoids](#), August 30
- [EM Asia FX/Rates Views: AI Exposure Driving Regional FX Disparity](#), August 29
- [Credit Notes: Expectations for Hyperscaler Debt Issuance: A Top-Down Approach](#), July 27
- [Global Markets Daily: Tracking Shifts in the Market's AI Views](#), August 25
- [Global Economics Comment: Is AI Impacting Global Labor Markets?](#), August 19
- [US Weekly Kickstart: What Q2 earnings reports signaled about the state of corporate AI adoption](#), August 14
- [India: AI Tracker: AI adoption and its impact on India's labour market](#), August 10
- [Global Economics Analyst: Assessing the Current Pace of AI Investment](#), August 2
- For research from prior quarters, see the [Artificial Intelligence](#) page on our publishing portal.

The Global Economics Team

Jan Hatzius

+1(212)902-0394
jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Joseph Briggs

+1(212)902-2163
joseph.briggs@gs.com
Goldman Sachs & Co. LLC

Sarah Dong

+1(212)357-9741
sarah.dong@gs.com
Goldman Sachs & Co. LLC

Megan Peters

+44(20)7051-2058
megan.l.peters@gs.com
Goldman Sachs International

Disclosure Appendix

Reg AC

We, Sarah Dong and Joseph Briggs, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Sarah Dong Goldman Sachs & Co. LLC, Joseph Briggs Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are

available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.